

Course Code : ECT 357

WYXZ/MS – 23 / 1601

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER NETWORKS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) Can you write an outline on TCP/IP reference model ? Briefly explain layer functionalities. 6(CO1)
 - (b) Can you make a distinction between connection-less and connection-oriented services used in Computer Communications ? 4(CO1)
2.
 - (a) What is purpose of transmission media in communication networks ? Briefly explain Unguided transmission media. 5(CO3)
 - (b) What is purpose of MAC protocols ? What could be done to minimize collisions in Pure Aloha ? 5(CO3)
3.
 - (a) A (7, 4) Hamming code is received as 1110000. Interpret the corrected code if even parity is used. How would you evaluate same code using Odd parity ? 5(CO4)
 - (b) Briefly explain need of flow control protocols in computer communications, explain Stop-and-Wait Protocol for noiseless channels. 5(CO4)
4.
 - (a) Justify how STS multiplexer of SONET differ from an AD multiplexer, when both can add signals together ? 5(CO4)

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Contd.

- (b) The addresses granted to an organization include 205.16.37.39/28 as one of its address. Estimate first, last and total number of addresses granted. 5(CO5)
5. (a) Illustrate your understanding of congestion control and explain any three open loop congestion control techniques. 5(CO4)
- (b) Write short notes on :—
- (i) Domain Name System.
- (ii) Digital Signature. 5(CO2)
6. What do you understand by cryptography in computer networks ? Explain in detail Symmetric key cryptography along with ciphers. What are the Shannon's Characteristics of "Good" Ciphers. 10(CO2)

